

Milestone 1

Dataset Description

The **VinDr-CXR** (VinBigData Chest X-ray) dataset was created for the automatic detection and localization of thoracic abnormalities in chest radiographs. It was released by VinBigData in collaboration with **Hospital 108** and **Hanoi Medical University Hospital** in Vietnam.

This dataset is unique because it provides **radiologist-level annotations** for both classification (presence of disease) and localization (bounding boxes), tackling the challenge of label noise and observer variability common in medical imaging.

2. Dataset Composition

The dataset consists of **18,000** postero-anterior (PA) view chest X-ray scans.

- **Training Set:** 15,000 images (Publicly labeled).
- **Test Set:** 3,000 images (Labels withheld for competition/benchmarking).
- **Image Format:** DICOM (.dicom or .dcm).
- **Bit Depth:** High dynamic range (often 12-bit or 16-bit), requiring normalization to 8-bit for standard computer vision models.

Training Data (15,000 images)

1. **Independent Labeling:** Each image in the training set was independently annotated by **3 experienced radiologists**.
2. **Result:** This leads to multiple annotations for the same image. For example, one radiologist might mark a "Nodule," while another might mark nothing or a different location.
3. **Implication for Models:** Models must handle noisy or conflicting ground truth labels (e.g., using Weighted Box Fusion or majority voting).

Test Data (3,000 images)

- **Consensus Labeling:** Each image in the test set was labeled by the **consensus of 5 radiologists** to ensure a high-quality "Gold Standard" for evaluation.

4. Class Labels and Statistics

The dataset focuses on **14 critical radiographic findings**. The train.csv file contains **67,914** total annotations across the 15,000 training images.

Class ID Map

Class ID	Class Name	Count (Train)	Description
0	Aortic enlargement	7,162	Abnormal widening of the aorta.
1	Atelectasis	279	Collapse of lung tissue.
2	Calcification	960	Calcium deposits in the lung.
3	Cardiomegaly	5,427	Enlarged heart.
4	Consolidation	556	Lung tissue filled with liquid instead of air.
5	ILD	1,000	Interstitial Lung Disease.
6	Infiltration	1,247	Substance denser than air (pus, blood) in lungs.
7	Lung Opacity	2,483	Opaque areas in the lung.
8	Nodule/Mass	2,580	Growths or lumps in the lung.
9	Other lesion	2,203	Lesions not fitting other categories.
10	Pleural effusion	2,476	Fluid build-up between lung layers.

11	Pleural thickening	4,842	Thickening of the pleura.
12	Pneumothorax	226	Air leaking into the space between lung and chest wall.
13	Pulmonary fibrosis	4,655	Scarring of the lung tissue.
14	No finding	31,818	No abnormalities detected.

Distribution Analysis

- **Dominant Class:** "No finding" (Class 14) is the most frequent, representing ~47% of all annotations.
- **Rare Classes:** *Pneumothorax* (226), *Atelectasis* (279), and *Consolidation* (556) are rare, making the dataset highly imbalanced. Techniques like oversampling or class-weighted loss functions are recommended.

5. File Structure and Metadata

train.csv Structure

The annotation file contains the following columns:

- `image_id`: Unique identifier for the scan.
- `class_name`: Human-readable name of the abnormality (e.g., "Cardiomegaly").
- `class_id`: Integer ID mapping to the class name (0-14).
- `rad_id`: Identifier of the radiologist who made the annotation (e.g., R10, R9). This allows filtering by specific annotators.
- `x_min`, `y_min`, `x_max`, `y_max`: Coordinates of the bounding box.
 - **Note:** For "No finding" (Class 14), these values are NaN.

Directory Structure

```
/kaggle/input/vinbigdata-chest-xray-abnormalities-detection/
|
├── train.csv          # Annotations for training images
├── sample_submission.csv # Format for submission
├── train/             # Folder containing 15,000 DICOM images
│   ├── [image_id].dicom
│   └── ...
├── test/              # Folder containing 3,000 DICOM images
│   ├── [image_id].dicom
│   └── ...
```

6. Preprocessing Requirements

As identified in the analysis notebook (image-conversion-and-annotation.ipynb), the raw data requires specific handling:

1. **DICOM Handling:** Files are in DICOM format, containing pixel data and metadata.
 - **Photometric Interpretation:** Some images may look inverted (MONOCHROME1 vs MONOCHROME2) and need to be corrected.
2. **Normalization:** Pixel arrays often contain high values (e.g., > 2000). To use standard CNNs (ResNet, EfficientNet, YOLO), images must be:
 - Normalized (min-max scaling).
 - Converted to 8-bit integers (0-255).
 - Saved as standard formats like PNG or JPG.
3. **Resize:** Original DICOMs are very high resolution (often 2000x2000+). Resizing to standard input sizes (e.g., 512x512 or 1024x1024) is necessary for training efficiency.

Dataset Preprocessing

1. Overview

This document details the preprocessing pipeline implemented to convert the raw **VinDr-CXR** DICOM dataset into a format compatible with YOLO object detection models. The pipeline handles image conversion, normalization, resizing, and label formatting.

2. Configuration Parameters

The following global parameters were used during the transformation process:

- **Target Image Size:** 640x640 pixels (Square resize for YOLO efficiency).
- **Seed:** 42 (Ensures reproducibility in data splitting).
- **Train/Val Split:** 80% Training / 20% Validation.
- **Exclusion:** Class 14 ("No finding") was removed from the label set to focus purely on object detection of abnormalities.

3. Image Processing Pipeline

The raw DICOM images required significant processing to be usable by standard computer vision libraries (OpenCV).

3.1 DICOM to PNG Conversion

The function `convert_dicom_to_png` performs the following operations:

4. **Photometric Correction:** Checks the `PhotometricInterpretation` tag. If the image is `MONOCHROME1` (where 0=White, 1=Black), the array is inverted to standard X-ray format (Bones=White, Air=Black).
5. **Normalization:** The high-bit depth DICOM pixel arrays are normalized to a 0-255 range using min-max scaling:

$$Pixel_{new} = \frac{Pixel - min}{max - min} \times 255$$

6. **Casting:** Converted to `uint8` format.

3.2 Resizing

- **Method:** Images are resized from their original resolution (approx. 2000x3000) to **640x640**.
- **Interpolation:** `cv2.INTER_AREA` is used, which is the preferred method for downscaling to prevent moiré patterns and preserve anatomical details.
- **Scaling Factor Calculation:** scaling factors are calculated for annotation adjustment:
 - `scale_x = 640 / original_width`

- $scale_y = 640 / original_height$

4. Annotation Processing

4.1 Filtering

The raw train.csv contains images labeled as "No finding" (Class ID 14).

- **Action:** Rows with `class_id == 14` were filtered out.
- **Rationale:** YOLO training requires bounding boxes for positive class instances. Images with no abnormalities contribute to the background class implicitly during training if no label file is provided (or empty file), but specific "No finding" boxes are not required.

4.2 Bounding Box Scaling

Since images were resized, bounding boxes were adjusted accordingly:

$$x_{min}^{new} = x_{min}^{old} \times scale_x$$

$$y_{min}^{new} = y_{min}^{old} \times scale_y$$

(Repeated for max coordinates)

4.3 Conversion to YOLO Format

The bounding boxes were converted from Pascal VOC format $[x_min, y_min, x_max, y_max]$ to normalized YOLO format $[x_center, y_center, width, height]$:

2. Calculate Center and Dimensions:

- $width = x_{max} - x_{min}$
- $height = y_{max} - y_{min}$
- $x_{center} = x_{min} + (width/2)$
- $y_{center} = y_{min} + (height/2)$

3. Normalize: Values are divided by the target image dimension (640).

- $x_{c_norm} = x_{center} / 640$
- $y_{c_norm} = y_{center} / 640$
- $w_{norm} = width / 640$

$$\circ \quad h_{norm} = height/640$$

5. Dataset Splitting & Organization

The processing loop iterated through unique image_ids to ensure no data leakage (all annotations for a single image stay in one split).

- **Training Set:** 80% of unique images.
- **Validation Set:** 20% of unique images.

Directory Structure Generated

The script automatically organized files into the standard YOLO directory structure:

```
/kaggle/working/vinbigdata-yolo-coco/
├── images/
│   ├── train/    # ~12,000 images (.png)
│   └── val/      # ~3,000 images (.png)
├── labels/
│   ├── train/    # Corresponding .txt files
│   └── val/      # Corresponding .txt files
└── data.yaml     # Configuration file for YOLO training
```

Label File Format (.txt)

Each text file corresponds to an image ID and contains one row per anomaly found:

```
<class_id> <x_center> <y_center> <width> <height>
0 0.4532 0.6781 0.1230 0.4421
3 0.1122 0.3321 0.1000 0.2000
```

6. Verification

A verification step was implemented using cv2.rectangle to draw the processed YOLO coordinates back onto the resized PNGs. This visual check confirmed that:

1. The "MONOCHROME1" inversion was successful (bones appear white).
2. The bounding boxes accurately encompass the anatomical abnormalities after resizing.